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Bone Conduction Microphone

Abstract

A bone conduction microphone is disclosed. The bone conduction microphone includes a circuit board, a housing engaged with the circuit board, a metal cover engaged with the circuit board for forming a first cavity, a vibration assembly, and a MEMS chip disposed in the first cavity. The metal cover includes a top wall and a side wall. The vibration assembly engages with the top wall for forming a second cavity and includes a vibration member and a spacer member. A first conduction cavity is formed by the housing, the vibration assembly, the metal cover and the circuit board. A vibration of the vibration member is transmitted to one side of the MEMS chip through the first conduction cavity, an acoustic channel of the circuit board and a back cavity of the MEMS chip. The bone conduction microphone reduces the electromagnetic interference and improves the SNR.

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Background/Summary

FIELD OF THE PRESENT DISCLOSURE

[0001] The present disclosure relates to acoustoelectric transducers, in particular to a bone conduction microphone.

DESCRIPTION OF THE RELATED ART

[0002] Bone conduction microphones convert slight vibrations of bones of the head and neck caused by human voice into electrical signals. Unlike traditional microphones collecting sounds through air conduction, bone conduction microphones can restore sounds with high definition even when used in noisy environments, so as to avoid the noise interference caused by airborne sounds and ensure the sound quality extremely high. However, a bone conduction microphone in the related art has an insufficient SNR (signal to noise ratio), because the electromagnetic interference affects a MEMS (micro electromechanical system) chip and an ASIC (application specific integrated circuit) chip that are used therein.

[0003] Thus, it is necessary to provide a novel bone conduction microphone to solve the problems.

SUMMARY

[0004] An objective of the present disclosure is to overcome the above problems and provide a bone conduction microphone which can reduce the electromagnetic interference and improve the SNR.

[0005] In order to achieve the objective mentioned above, the present disclosure discloses a bone conduction microphone including a circuit board having an acoustic channel, a housing engaged with the circuit board for forming a containing space, a metal cover disposed in the containing space and engaged with the circuit board for forming a first cavity, a vibration assembly disposed in the containing space, a first conduction cavity, and a MEMS chip disposed in the first cavity and fixed to the circuit board. The metal cover includes a top wall spaced apart from the circuit board and a side wall disposed between the top wall and the circuit board. The vibration assembly engages with the top wall for forming a second cavity and includes a vibration member spaced apart from the top wall and a spacer member disposed between the vibration member and the top wall. The first conduction cavity is formed by the housing, the vibration assembly, the metal cover and the circuit board. The MEMS chip has a back cavity, and the acoustic channel communicates the first conduction cavity and the back cavity. A vibration of the vibration member is transmitted to one side of the MEMS chip through the first conduction cavity, the acoustic channel and the back cavity.

[0006] As an improvement, the top wall is provided with a communication hole communicating the first cavity and the second cavity. A second conduction cavity is formed by the first cavity, the communication hole and the second cavity. The vibration of the vibration member is also transmitted to another side of the MEMS chip through the second conduction cavity.

[0007] As an improvement, the top wall is a metal plate with a porous structure. The porous structure is arranged in an array, and the communication hole is the porous structure.

[0008] As an improvement, the top wall is a metal mesh, and the communication hole is a mesh hole of the metal mesh.

[0009] As an improvement, the metal cover is formed integrally, or the metal cover is formed by

combining the top wall and the side wall.

[0010] As an improvement, the vibration member includes a vibration membrane fixed to the spacer member and a mass block fixed to the vibration membrane.

[0011] As an improvement, the bone conduction microphone further includes an ASIC chip electrically connected to the MEMS chip. The ASIC chip is arranged in the first cavity and fixed to the circuit board.

[0012] As an improvement, the housing is made of metal.

[0013] As an improvement, the spacer member is a part of the top wall, and a protrusion of the top wall extending in a direction away from the circuit board forms the spacer member.

[0014] As an improvement, the spacer member is a part of the vibration membrane, and a protrusion of the vibration membrane extending in a direction close to the circuit board forms the spacer member.

[0015] As an improvement, the vibration membrane is an air-permeable membrane.

[0016] As an improvement, the MEMS chip includes a vibration diaphragm, and the vibration diaphragm is an air-permeable diaphragm.

[0017] In the bone conduction microphone according to the present disclosure, the metal cover has the function of electromagnetic shielding, so as to reduce the electromagnetic interference affecting the MEMS chip and the ASIC chip, and improve the SNR of the bone conduction microphone. In addition, the top wall of the metal cover can protect the MEMS chip and the ASIC chip from the impact of the vibration member, thereby improving the reliability of the bone conduction microphone.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] In order to more clearly illustrate the technical solutions in embodiments of the present disclosure, the accompanying drawings used in the description of the embodiments will be briefly introduced below. It is apparent that, the accompanying drawings in the following description are only some embodiments of the present disclosure, and other drawings can be obtained by those of ordinary skill in the art based on the accompanying drawings without creative efforts, wherein:

[0019] FIG. 1 is a structural schematic diagram of a bone conduction microphone in accordance with an exemplary embodiment of the present disclosure.

[0020] FIG. 2 is a structural schematic diagram of a top wall of a metal cover of the bone conduction microphone in FIG. 1.

[0021] FIG. 3 is a structural schematic diagram of another bone conduction microphone of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0022] The technical solutions in embodiments of the present disclosure will be described clearly and completely below with reference to the accompanying drawings in the embodiments of the present disclosure. It is apparent that, the described embodiments are merely some of rather than all of the embodiments of the present disclosure. All other embodiments acquired by those of ordinary skill in the art without creative efforts based on the embodiments of the present disclosure shall fall within the protection scope of the present disclosure.

[0023] Referring to FIG. 1, the present disclosure discloses a bone conduction microphone **100**. The bone conduction microphone **100** includes a circuit board **1** having an acoustic channel **10**, a housing **2** engaged with the circuit board **1** for forming a containing space, a metal cover **3** disposed in the containing space and engaged with the circuit board **1** for forming a first cavity **30**, a vibration assembly **4** disposed in the containing space, and a MEMS chip **5** with a back cavity **50** disposed in the first cavity **30** and fixed to the circuit board **1**.

[0024] The metal cover **3** includes a top wall **31** spaced apart from the circuit board **1** and a side wall **32** disposed between the top wall **31** and the circuit board **1**.

[0025] The vibration assembly **4** includes a vibration member **41** spaced apart from the top wall **31** and a spacer member **42** disposed between the vibration member **41** and the top wall **31**. The vibration assembly **4** engages with the top wall **31** for forming a second cavity **40**.

[0026] A first conduction cavity **101** is formed by the housing **2**, the vibration assembly **4**, the metal cover **3** and the circuit board **1**. The acoustic channel **10** communicates the first conduction cavity **101** and the back cavity **50**. A vibration of the vibration member **41** is transmitted to one side of the MEMS chip **5** through the first conduction cavity **101**, the acoustic channel **10** and the back cavity **50**.

[0027] The vibration member **41** includes a vibration membrane **411** fixed to the spacer member **42** and a mass block **412** fixed to the vibration membrane **411**. In this embodiment, the mass block **412** is fixed to a lower side of the vibration membrane **411**. In another embodiment as shown in FIG. 3, the mass block **412** can be fixed to an upper side of the vibration membrane **411**. Optionally, the vibration membrane **411** can be an air-permeable membrane, the air-permeable membrane can be made of air-permeable material, or the air-permeable membrane has an air-permeable structure including but not limited to the hole, gap, notch and movable valve.

[0028] The bone conduction microphone **100** further includes an ASIC chip **6** electrically connected to the MEMS chip **5**. The ASIC chip **6** is arranged in the first cavity **30** and fixed to the circuit board **1**. In other embodiments, the ASIC chip **6** can be fixed to the metal cover **3**, or integrated with the MEMS chip **5**.

[0029] The metal cover **3** has the function of electromagnetic shielding, so as to reduce the electromagnetic interference affecting the MEMS chip **5** and the ASIC chip **6**, and improve the SNR of the bone conduction microphone **100**. In addition, the top wall **31** of the metal cover **3** can protect the MEMS chip **5** and the ASIC chip **6** from the impact of the vibration member **41**, thereby improving the reliability of the bone conduction microphone **100**.

[0030] Optionally, the housing **2** is made of metal, so as to further improve the effect of electromagnetic shielding and the overall structural strength of the bone conduction microphone **100**.

[0031] Optionally, the metal cover **3** can be formed integrally, or the metal cover **3** is formed by combining the top wall **31** and the side wall **32**. The top wall **31** and the side wall **32** are welded or glued together.

[0032] Optionally, the top wall **31** can be a wall without any hole, or, the top wall **31** is provided with a communication hole **311** communicating the first cavity **30** and the second cavity **40**, in this embodiment, a second conduction cavity **102** is formed by the first cavity **30**, the communication hole **311** and the second cavity **40**. The vibration of the vibration member **41** is also transmitted to another side of the MEMS chip **5** through the second conduction cavity **102**, thereby improving the sensitivity of the bone conduction microphone **100** by differential means. In the embodiment as shown in FIG. 2, the top wall **31** is a metal plate with a porous structure **3111**. The porous structure **3111** is arranged in an array, and the communication hole **311** is the porous structure **3111**. In other embodiments, the top wall **31** can be a metal mesh, and the communication hole **311** is a mesh hole of the metal mesh. It should be noted that, the communication hole **311** can be a single hole, double holes, or multiple holes more than two, in addition, the number, position, shape, size, formation manner and arrangement manner of the communicating hole **311** are not limited, as long as the communicating hole **311** can communicate the first cavity **30** and the second cavity **40**.

[0033] Optionally, the MEMS chip **5** includes a vibration diaphragm **51**, and the vibration diaphragm **51** can be an air-permeable diaphragm, the air-permeable diaphragm can be made of air-permeable material, or the air-permeable diaphragm has an air-permeable structure including but not limited to the hole, gap, notch and movable valve.

[0034] Optionally, the spacer member **42** can be an independent frame structure, or, the spacer

member **42** is a part of the top wall **31**, and a protrusion of the top wall **31** extending in a direction away from the circuit board **1** forms the spacer member **42**, or, the spacer member **42** is a part of the vibration membrane **411**, and a protrusion of the vibration membrane **411** extending in a direction close to the circuit board **1** forms the spacer member **42**.

[0035] The above are only embodiments of the present disclosure. It should be pointed out that those of ordinary skill in the art may also make improvements without departing from the ideas of the present disclosure, all of which fall within the protection scope of the present disclosure.

Claims

1. A bone conduction microphone, comprising: a circuit board, having an acoustic channel; a housing, engaged with the circuit board for forming a containing space; a metal cover, disposed in the containing space and engaged with the circuit board for forming a first cavity, comprising a top wall spaced apart from the circuit board and a side wall disposed between the top wall and the circuit board; a vibration assembly, disposed in the containing space and engaged with the top wall for forming a second cavity, comprising a vibration member spaced apart from the top wall and a spacer member disposed between the vibration member and the top wall; a first conduction cavity, formed by the housing, the vibration assembly, the metal cover and the circuit board; and a MEMS chip, disposed in the first cavity and fixed to the circuit board, having a back cavity, wherein the acoustic channel communicates the first conduction cavity and the back cavity, a vibration of the vibration member is transmitted to one side of the MEMS chip through the first conduction cavity, the acoustic channel and the back cavity.
2. The bone conduction microphone as described in claim 1, wherein the top wall is provided with a communication hole communicating the first cavity and the second cavity, a second conduction cavity is formed by the first cavity, the communication hole and the second cavity, the vibration of the vibration member is also transmitted to another side of the MEMS chip through the second conduction cavity.
3. The bone conduction microphone as described in claim 2, wherein the top wall is a metal plate with a porous structure, the porous structure is arranged in an array, the communication hole is the porous structure.
4. The bone conduction microphone as described in claim 2, wherein the top wall is a metal mesh, the communication hole is a mesh hole of the metal mesh.
5. The bone conduction microphone as described in claim 1, wherein the metal cover is formed integrally, or the metal cover is formed by combining the top wall and the side wall.
6. The bone conduction microphone as described in claim 1, wherein the vibration member comprises a vibration membrane fixed to the spacer member and a mass block fixed to the vibration membrane.
7. The bone conduction microphone as described in claim 1, further comprising an ASIC chip electrically connected to the MEMS chip, the ASIC chip is arranged in the first cavity and fixed to the circuit board.
8. The bone conduction microphone as described in claim 1, wherein the housing is made of metal.
9. The bone conduction microphone as described in claim 1, wherein the spacer member is a part of the top wall, a protrusion of the top wall extending in a direction away from the circuit board forms the spacer member.
10. The bone conduction microphone as described in claim 6, wherein the spacer member is a part of the vibration membrane, a protrusion of the vibration membrane extending in a direction close to the circuit board forms the spacer member.
11. The bone conduction microphone as described in claim 6, wherein the vibration membrane is an air-permeable membrane.

12. The bone conduction microphone as described in claim 1, wherein the MEMS chip comprises a vibration diaphragm, the vibration diaphragm is an air-permeable diaphragm.
